Postdoc/Research Associate (all genders) - Social Learning and Cultural Evolution

Prof. Charley Wu invites applications for a Research Associate (Postdoc; 3 years of funding at 100% TV-E14) to join his group at TU Darmstadt. The position is co-advised by **Prof. Dominik Deffner** (University of Marburg), and is embedded within the Cluster of Excellence "The Adaptive Mind" (TAM), to which both PIs belong.

We are explicitly seeking an experimentalist with demonstrated expertise in designing, implementing, and analyzing behavioral experiments, particularly multiplayer online experiments and/or immersive, in-person laboratory experiments (e.g., Wu et al., 2025; Deffner et al., 2024).

The successful candidate will play a central role in developing a new generation of large-scale behavioral studies on social learning and cultural evolution. This includes online multiplayer paradigms (e.g., interactive transmission chains, networked coordination tasks, real-time social interaction experiments), as well as in-person, immersive experiments using a newly constructed state-of-the-art tracking hall in Darmstadt that enables high-resolution behavioral tracking and interactive group studies.

The deadline is March 30th, 2026 or until filled.

Research Environment

Prof. Wu (W3 Professor for Computational Cognitive Science) leads the "Human and Machine Cognition" lab, situated at the intersection of cognitive science and machine learning. The lab integrates rigorous theoretical frameworks with computational modeling approaches to investigate human cognition, learning, decision-making, social interaction, and cultural evolution. These methods include Bayesian modeling, reinforcement learning, program induction, and information theory. Prof. Wu's research is funded by Hessian AI, an ERC Starting Grant, and a LOEWE Start Professorship. For more information on Prof. Wu's research group, see hmc-lab.com

Prof. Deffner is Assistant Professor for Computational Modelling of Behaviour at the Department of Psychology. His work combines dynamic group experiments, (cultural) evolution, and field research with statistics and computational modelling. The lab investigates the cognitive processes and social as well as cultural dynamics that enable individuals and collectives to adapt to changing and uncertain environments. Moreover, his work develops new statistical methods and causal-inference workflows, especially for cross-cultural research. For more information, see https://www.uni-marburg.de/en/fb04/team-deffner

Positions will be based at TU Darmstadt, Germany, with flexible starting dates. Salaries are competitive by U.S. and European standards and commensurate with experience and expertise. The working language of the lab and broader academic community is English; fluency in German is not required, though the university provides free German language courses for interested scientific staff. The position is embedded within the Excellence Cluster The Adaptive Mind (TAM), a large interdisciplinary research network spanning cognitive science, psychology, neuroscience, behavioral economics, and artificial intelligence. TAM provides a highly collaborative, well-funded environment with strong international visibility and extensive opportunities for interdisciplinary exchange.

Your tasks:

In particular, the tasks include:

- Design, implement, and lead experimental research programs on social learning and cultural evolution
- Develop and run multiplayer online behavioral experiments, including real-time interactive paradigms
- Develop and conduct immersive, in-person group experiments in a new high-resolution behavioral tracking hall in Darmstadt

- Integrate experimental data with computational modeling approaches (e.g., Bayesian inference, reinforcement learning, information-theoretic analyses)
- Supervise and collaborate with PhD students, particularly on experimental design and data collection
- Contribute to grant writing and collaborative projects within TAM
- Publish in leading international journals and present at major conferences
- Actively engage with interdisciplinary research networks within cognitive science, AI, and the TAM cluster

Your profile:

- A PhD in Cognitive Science, Psychology, Behavioral Economics, Computational Neuroscience, Artificial Intelligence, or a closely related discipline
- Strong, demonstrated expertise in experimental methods, particularly interactive, multi-participant, or immersive behavioral paradigms
- Experience designing and running online multiplayer experiments (e.g., oTree, jsPsych, custom JavaScript platforms, Prolific-based studies, networked experiments) and/or laboratory-based group interaction studies
- Extensive experience with quantitative data analysis and computational modeling
- Strong programming skills (e.g., Python, R, MATLAB, JavaScript)
- A strong publication record relative to career stage
- Demonstrated scientific independence and leadership potential
- Excellent written and spoken English

We offer:

TU Darmstadt offers varied and challenging assignments, freedom to work independently, the latest technologies, good collaboration between colleagues in partnership, needs-based training opportunities and customized personnel development.

The fulfillment of the duties likewise enables the scientific qualifications of the candidate.

- Development and organisation – comprehensive in-house training offers, including the opportunity for continuing education and development;
- Annual leave/educational leave – 30 days annual leave (full-time employment) and 5 days educational leave;
- Sustainable and mobile – eligibility to free public transport in the state of Hesse with the LandesTicket Hessen (Hesse StateTicket) in accordance with the currently valid collective agreement, in addition to opportunities to working mobile at times;
- Fit and healthy – free of charge preventive medical check-ups and a wide-ranging subsidised sports programme
- Work-life balance – flexible working time models, plus BGM (*Betriebliches Gesundheitsmanagement* – University Health Management);
- Pension scheme – supplementary public service pension scheme (VBL) in accordance with the currently applicable regulations;
- University bicycle
- Family-friendliness/compatibility of family/care/career – (university-run) childcare services, child allowance (based on the collective agreement), childcare programmes during school holidays

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About us:
TU Darmstadt stands for excellent and relevant science. We shape far-reaching processes of global change—from energy transition to artificial intelligence—through outstanding scientific knowledge and innovative academic programs. Our cutting-edge research focuses on three fields: Energy and Environment, Information and Intelligence, Matter and Materials. With strong ties to the Frankfurt Rhine-Main metropolitan region (~18 mins to Frankfurt HBF by train), we have an exceptionally international orientation and actively support European integration.

About our department:
TU Darmstadt is one of Europe's leading institutions in cognitive science and artificial intelligence (csrankings.org), bringing together interdisciplinary research on cognition through the Centre for Cognitive Science and intelligent systems as a member of the European Laboratory for Learning and Intelligent Systems (ELLIS). The department provides a vibrant, inclusive research environment and encourages extensive collaboration with leading cognitive science and AI researchers, both locally (via the Hessian AI Center) and internationally.

Contact:

If you have any questions about this position, please contact Charley Wu (charley[dot]wu[at]tu-darmstadt[dot]de).

Your application should be written in **English** and include a cover letter, CV, research statement (1-2 pages), and contact info for 2 references. Your **research statement** (1-2 pages) should describe your research interests, expertise, and an explanation for why you are a particularly good fit for the position (references to prior published research and links to public code repositories are appreciated).

All application materials should be submitted in a single PDF to Derya Pullum (pullum[at]psychologie[dot]tu-darmstadt]dot]de) **with the subject line** "Postdoc application: Social Learning and Cultural Evolution"